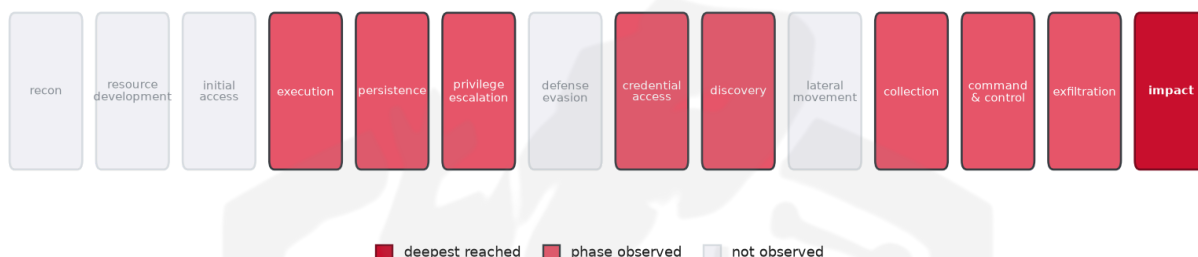


THREAT REPORT: VOID ARACHNE DEPLOYS VALLEYRAT VIA SHORTCUT MODIFICATION

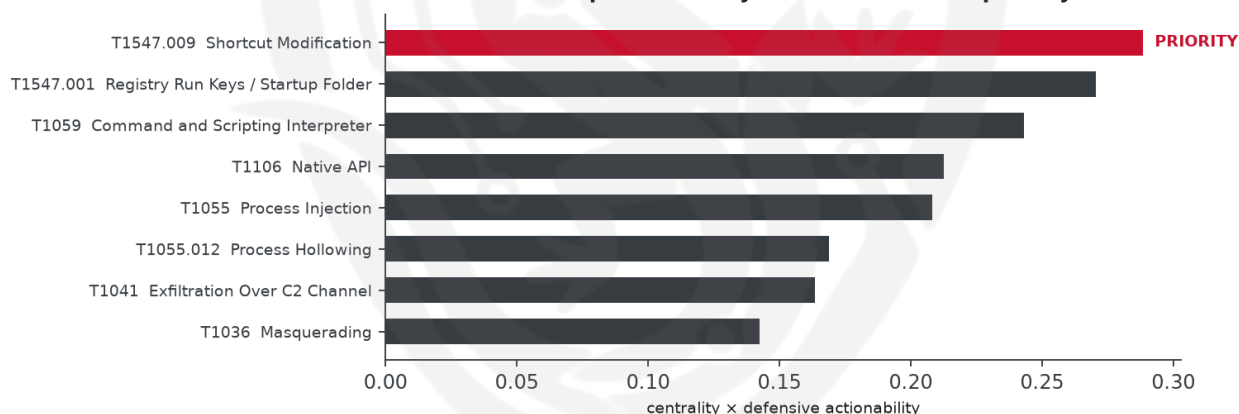
Visual Summary

Kill-Chain Reach — Critical (deepest phase reached: impact)



How far down the attack chain this campaign reached; deepest phase in crimson.

Defensive Chokepoint — why this control is the priority



The control that most disrupts the attack (priority in crimson).

Summary

Void Arachne is observed distributing the ValleyRAT remote-access trojan, masquerading it as adware. The campaign targets systems in the British Indian Ocean Territory, China, and India, though no specific CVE is cited. A range of credential-stealing, system-discovery, and exfiltration techniques are employed. Priority should be given to baselining and monitoring autostart locations (Run keys, startup folders) and alerting on any new persistence entries.

Threat Overview

The threat actor identified as Void Arachne leverages the ValleyRAT malware family. No concrete vulnerability is referenced in the source (central CVE marked as Data Insufficient). The activity focuses on victims located in the British Indian Ocean Territory, China, and India, with sector information not provided.

Attack Chain and Priority Control

The observed operations begin with masquerading the payload as adware, followed by execution via shortcut modification (T1547.009). Once active, the malware performs system checks, gathers system and user information, captures screen and clipboard data, logs keystrokes, and injects into legitimate processes (including process hollowing). It then establishes a web-based command-and-script interpreter, transfers additional tools, and exfiltrates data over its C2 channel. Persistence is reinforced through shortcut modification and registry run keys/startup folders, while defenses are impaired by disabling security tools and, if needed, shutting down or rebooting the system.

Priority technique: T1547.009 Shortcut Modification

Priority control: Baseline and monitor autostart locations (Run keys, startup folders); alert on new persistence entries.

Infrastructure and Corroboration

Abuse-ch corroborates that the malicious payload belongs to the ValleyRAT family. The command-and-control servers are hosted by CTG Server Limited, as observed in the associated ASN data. No indicators reached the 80 % confidence threshold on AbuseIPDB, and no high-confidence reputation flags were reported.

Technical Appendix

Ground-truth telemetry from the source pulse; analytical scores are secondary and clearly labelled.

Observed Techniques (ATT&CK, expert-tagged by source)

- T1113 Screen Capture
- T1033 System Owner/User Discovery
- T1056.001 Keylogging
- T1497.001 System Checks
- T1115 Clipboard Data
- T1082 System Information Discovery
- T1106 Native API
- T1036 Masquerading
- T1055 Process Injection
- T1547.009 Shortcut Modification
- T1059 Command and Scripting Interpreter
- T1041 Exfiltration Over C2 Channel
- T1547.001 Registry Run Keys / Startup Folder
- T1562.001 Impair Defenses: Disable or Modify Tools
- T1055.012 Process Hollowing
- T1027 Obfuscated Files or Information
- T1071.001 Web Protocols

- T1574.002 Hijack Execution Flow
- T1105 Ingress Tool Transfer
- T1529 System Shutdown/Reboot

Analytical Scores

- Priority technique (graph chokepoint): T1547.009 Shortcut Modification (centrality 0.3037).
- Operational risk: Critical (score 1.0, reaches impact).

Behavioral Signature Cluster

- Method: technique-overlap cosine vs 170 MITRE ATT&CK Groups (top overlap 0.4286; reference threshold $\tau = 0.6$).
- These are behavioural resemblances for defensive playbook cross-referencing, not an identity assessment. Where the source pulse names an actor, that attribution is authoritative; the overlaps below neither confirm nor contest it.

Nearest historical profiles by behavioural overlap (resemblance only):

Historical Profile	Behavioural Overlap
APT37	0.4286
Windshift	0.4256
Darkhotel	0.3934
APT18	0.3932
APT39	0.3686

Enrichment Signal

- Highest AbuseIPDB confidence among IOCs: 0%.
- Malware families corroborated by abuse.ch: ValleyRAT.
- Note: enrichment raises the severity signal (an IOC at or above 80% AbuseIPDB confidence, or a confirmed malware-family hit). This is context, not a change to the source assessment.

Indicators of Compromise (defanged — non-clickable)

The network and file indicators observed in this activity are listed below for blocklisting:

Type	Indicator	Context
IP	103[.]45[.]66[.]18	AbuseIPDB 0% (HK) · ValleyRAT · AS152194 CTG Server Limited

Type	Indicator	Context
IP	192[.]253[.]225[.]173	AbuseIPDB 0% (HK) · ValleyRAT · AS152194 CTG Server Limited
Hash	c24e99f9437feacaa63766a3cde3fe3d	ValleyRAT
Hash	07ddbbe2c71c45577a7a4fbcdba0df91	ValleyRAT
Hash	8a626d844943da3456b044f38deae3a2	ValleyRAT

Reputation by AbuseIPDB · malware attribution by abuse.ch · Geo/ASN data by IP2Location LITE.

Flame Cell // Adversary Pivot [BETA]

BETA. A hypothetical tabletop projection, anchored to documented ATT&CK behaviour and technique co-occurrence across 170 tracked groups. It is not a prediction; treat it as a war-game prompt and review before acting. Spot a pivot that looks wrong? norbert@salacti.com

- **Control applied:** T1547.009 Shortcut Modification (persistence)
- **Observed here:** T1547.009 Shortcut Modification was the only persistence technique observed in this campaign, so the pivot is projected from cross-actor behaviour.
- **Projected pivot:** T1053 Scheduled Task/Job (persistence)
- **Basis:** of the 37+ groups that use Shortcut Modification, this pairing runs 1.9x above base rate, a disproportionately-coupled move rather than the obvious one.

With T1547.009 Shortcut Modification blocked, Void Arachne could preserve the same objective by pivoting to T1053 Scheduled Task/Job, a technique disproportionately paired with it across tracked groups. Defensive countermeasure: Alert on new scheduled tasks (Event ID 4698); restrict schtasks/at to admins; baseline legitimate task creators.

Also plausible (same tactic): T1543 Create or Modify System Process (n=18, lift 1.8), T1098 Account Manipulation (n=12, lift 2.0)

Detection and hardening for the pivot (T1053 Scheduled Task/Job)

- **Telemetry:** Windows Security (Object Access audit); Microsoft-Windows-TaskScheduler/Operational; Sysmon
- **Detect:**
 - Security 4698 (task created), 4702 (updated), 4700/4701 (enabled/disabled), 4699 (deleted)
 - TaskScheduler/Operational 106 (registered), 140 (updated), 141 (deleted)
 - Sysmon 1 for schtasks.exe and at.exe; Sysmon 11 for writes under C:\Windows\System32\Tasks\
- **Harden:**

- Restrict schtasks/at to administrators; disable the legacy at.exe
- Baseline the legitimate task creators and alert on anything outside it
- **MITRE:** M1047 Audit, M1028 Operating System Configuration, M1026 Privileged Account Management · DS0003 Scheduled Job, DS0009 Process, DS0022 File

